TOR 7 – Research Recommendations

Review, evaluate, and report on the status of research recommendations from the last assessment peer review, including recommendations provided by the prior assessment working group, peer review panel, and SSC. Identify new recommendations for future research, data collection, and assessment methodology. If any ecosystem influences from TOR 1 could not be considered quantitatively under that or other TORs, describe next steps for development, testing, and review of quantitative relationships and how they could best inform assessments. Prioritize research recommendations.

1. Previous Research Recommendations

The working group reviewed research recommendations from the 2018 benchmark stock assessment and peer review in (SAW/SARC 65); the 2020 management track assessment and peer-review, and the New England Fishery Management Council's Science and Statistical Committee (SSC) meetings in 2020, 2021, 2022, 2023, and 2024. The following is a catalogue of research recommendations from the most recent stock assessments in 2020 and SSC input. The working group notes that not all SSC comments and recommendations are within the purview of the working group. Since there is substantial overlap in the themes of research recommendations, Section 2 of this working paper addresses the status of these recommendations by topic/theme.

2018 Benchmark Recommendations

1. Further investigate methods for better survey coordination between the various survey programs, including survey design, timing, and standardized data formatting for easier sharing.
2. Investigate changes in dredge efficiency and saturation due to high scallop densities or high bycatch rates.
3. Analyze past juvenile scallop mortality events and develop better methods to model time-varying mortality in the assessment models.
4. Collect information needed for the management of the GOM fishery and development of appropriate reference points including biological parameters, fishery-independent surveys, and fishery-dependent data.
5. Continue development of scallop ageing methods and examination of scallop growth processes including density dependent effects.
6. Improve training of annotators used in optical surveys and develop standardized QA/QC procedures for data collected from imagery.
7. Investigate use of software for automated annotation of imagery from optical surveys.

8. Investigate methods to better estimating biomass and abundance variances from HabCam optical surveys including development of Bayesian geostatistical methods.
9. Investigate and estimate current and historical unreported landings and effects of spatially heterogeneous fishing mortality on mortality estimates.
10. Develop a spatially-explicit methodology for forecasting the abundance and distribution of sea scallops by incorporating spatial data from surveys, landings, and fleet effort (aka GEOSAMS).
11. Investigate and parameterize sub-lethal effects of disease, parasites, or discarding on mortality, growth, and landings.
12. Revive and streamline previously-developed methods for interpreting VMS data.
13. Further refine and test methods for forecasting LPUE.
14. Continued investigation of discard mortality, particularly during warm water periods, by incorporating environmental data.
15. Continue improvements of observer recordings for vessel fishing behavior including deck loading and shucking dynamics in responses to disease or poor scallop health.
16. Continue investigating the extent of incidental fishing mortality, particularly on hard bottom habitats.

2020 Management Track Assessment

Examine different growth rates in different harvesting areas, particularly the Nantucket Lightship region.

Further work to develop gonad-based estimates of SSB and reference points.

Explore runs of previous assessment model configurations to compare to the new version of the assessment.

Additionally, the Review Panel was asked to evaluate a new approach being developed in the SYM model that models selectivity dynamically as a function of full recruitment fishing mortality. It was demonstrated that this dynamic selectivity model better characterizes selectivity as fishing mortality moves towards more extreme values and can result in more precise estimates of biological reference points when taken into account. The Peer Review Panel encouraged continued development of this approach and while it will not be used in this upcoming assessment cycle we look forward to its full implementation at a later date.

2020 SSC Report

The SSC notes that the SAMS model seems to be having some difficulty capturing some of the recent stock changes. The SSC does understand and supports the need to make the adjustments noted above in an effort to produce the most reliable model results for setting catch advice, but spending time developing upgrades to the existing modeling framework seem needed at this point. The SSC felt this could happen on two tracks. A review of the SAMS forecasting model may be achievable under the new management track assessment process by upgrading the scallop assessment to a level 2 or 3 review. One explicit thing discussed by the SSC was looking into new treatments for recruitment assumptions in the SAMS model as an item that potentially could be addressed during a management track assessment in an enhanced review level. In parallel, a revamp of the SAMS model to allow for more spatial estimation would be another fruitful area to explore. The SSC recognizes that this would have to wait for the 2024 research track scheduled for this species. In summary, the SSC recommends a review of the SAMS model in the next management track assessment, and supports NEFSC's development of a geostatistical SAMS model for the 2024 research track assessment

2021 SSC Report

The SSC recommends that in the future a more holistic consideration of stock changes and a more systematic approach to adjusting survey and model parameters be employed.

The SSC recommends consideration of the future of surveys in the GOM region be included in the ongoing Scallop Survey Working Group and NEFSC-supported scallop survey re-stratification efforts.

The SSC recommends that ongoing research on potential drivers of changes in sea scallop stock dynamics (e.g., changing ocean conditions, including ocean acidification and warming) be included in the upcoming review of the SAMS model and in the 2024 research track assessment for scallops.

2022 SSC Report

The SSC recommends continued evaluation of the performance of the projection model and the need for a more holistic evaluation of changes in stock dynamics, a synoptic evaluation of potential drivers (e.g., changing ocean conditions, including ocean warming), and revision of model assumptions to account for these changes. The SSC strongly recommends integration of the results of ongoing research examining drivers of changing stock dynamics in the upcoming Research Track assessment for scallops.

A Research Track assessment is planned for scallops in 2024. The SSC recommends investigation of the environmental drivers affecting the productivity of this stock, such as those influencing recruitment and natural mortality. The SSC requests an opportunity to provide feedback on the Terms of Reference for the Scallop Research Track assessment when available.

2023 SSC Report

The SSC has serious concerns about increased natural mortality in the Virginia and Delmarva areas, which NOAA Fisheries staff suggested was linked to increases in bottom temperature in that region. Given the high value of this fishery it is critical to understand the implications of expected future increases in bottom temperature on natural mortality and how these impacts on scallop survival will likely continue to extend northward. The SSC suggests that a thorough examination of the impacts of ecosystem and climate change on population dynamics is critical during the ongoing Sea Scallop Research Track Stock Assessment. Further, a description of these observations should be included in the Mid-Atlantic State of the Ecosystem report. The scallop population was previously identified by NEFSC scientists as highly vulnerable to climate change (Hare et al. 2016), and the assessment analyst noted the prospective synergistic adverse effects of temperature and acidification on the stock. It is imperative to translate this known vulnerability into an action plan for this fishery. In addition, the sedentary nature of scallops combined with the variety of data streams available (e.g., dredge, HabCam, drop camera) make this an ideal case study to examine the implications of climate change on northeast US fisheries and to help plan for potential impacts on other species that have dynamics that are less well understood.

2024 SSC Report

The SSC supports continued monitoring of changes in the dynamics of this stock (i.e., recruitment, growth, and natural mortality) and research to understand the role of environmental drivers as this represents the most pressing concern facing the future of this fishery. NEFSC staff and the PDT have been monitoring this issue and it is being examined in the Research Track Assessment. The SSC also suggests a retrospective-type analysis on historical area-specific SAMS projections; a quantitative measure of past bias could be used as a tool to inform future projections. Transitioning the CASA model into the "next generation" of assessments would also be a natural progression; a state-space model with parameters that can be connected to environmental variables could result in more accurate estimates of biomass and fishing mortality rate and reference points with an improved understanding of uncertainty and the relationship between scallops and the ecosystem. Additional sampling or analyses of Gulf of Maine scallops would obviate the need for Georges Bank proxies of FMSY and growth parameters. The inclusion of more comprehensive socioeconomic data as context for SSC decision making would be helpful to better understand the social implications of the suggested OFLs and ABCs. Finally, the Council should consider more formal use of risk tables to characterize ecosystem considerations around bycatch risk in the area management decision making process.

2. Response to Research Recommendations

The following summarizes the working group's response to previous research recommendations. Bulleted items are the research recommendations, and the sub-bullets are the working group's responses:

<u>Surveys:</u>

- Further investigate methods for better survey coordination between the various survey programs, including survey design, timing, and standardized data formatting for easier sharing. (2018 Benchmark Assessment)
    - The NEMFC and Northeast Fisheries Science Center (NEFSC) formed a Scallop Survey Working Group (SSWG) to address these topics. The SSWG developed recommendations to address four Terms of Reference, including survey spatial coverage, sampling intensity and frequency, data standardization, storage, and access, potential impacts from the development of offshore wind, and data needs to support future stock assessments.[1]
- Investigate changes in dredge efficiency and saturation due to high scallop densities or high bycatch rates. (2018 Benchmark Assessment)
    - Ongoing research at the Virginia Institute of Marine Science. Not completed.
- Collect information needed for the management of the Northern Gulf of Maine (NGOM) fishery and development of appropriate reference points including biological parameters, fishery-independent surveys, and fishery-dependent data. (2018 Benchmark Assessment)
    - Members of the research track working group assembled a NGOM working paper, which includes updated survey and fishery data. The survey-time series continues to grow with continued Research Set-Aside

[1] More information at: *https://d23h0vhsm26o6d.cloudfront.net/1.-SSWG-Report-September-2022.pdf*

support. Additional data is needed to develop reference points for this region.

- Improve training of annotators used in optical surveys and develop standardized QA/QC procedures for data collected from imagery. (2018 Benchmark Assessment)
    - o Standardized QA/QC procedures have been developed for NEFSC HabCam, along with a detailed training and annotation manual that were implemented in 2022. SMAST Drop Camera program also uses standardized training sets, and a QA/QC process. Substantial progress has been made.
- Investigate the use of software for automated annotation of imagery from optical surveys. (2018 Benchmark Assessment)
    - o Work is underway at NEFSC. AI algorithms continue to improve rapidly. NEFSC staff have submitted a publication in this area (focused on sand dollars), though staff capacity has been a challenge to achieve full implementation.
    - o Other survey groups that work with optical tools (SMAST Drop Cam and Coonamessett Farm Foundation HabCam) have ongoing projects in this area.
    - o A similar recommendation is made by the working group for future research recommendations.
- Consideration of the future of surveys in the GOM region be included in the ongoing Scallop Survey Working Group (SSWG) and NEFSC-supported scallop survey re-stratification efforts. (2021 SSC Report)
    - o A sub-group of the SSC reviewed and recommended the use of Generalized Random Tessellation Stratified (GRTS) method for survey design for Georges Bank and the Mid-Atlantic. Future implementation could be done in the Gulf of Maine. The sub-group report is available at this page: https://www.nefmc.org/library/september-2024-ssc-report.
    - o There are two recommendations from the SSWG report that are particularly relevant to surveys of the Gulf of Maine region. First, the Northern Gulf of Maine management area and Gulf of Maine resource area should be included in regular survey coverage. Second, the effort should be made to match appropriate sampling tools, designs, and methods with specific conditions of survey areas. The Gulf of Maine is heavily fished using fixed gear (e.g., lobster pots with vertical lines), which presents challenges for sampling with mobile gear (either dredge, or a towed camera system). Of the two mobile sampling techniques, dredge

sampling is generally towed over shorter distances than HabCam, which runs transects over larger areas on Georges Bank and the Mid-Atlantic. There is no dedicated federal funding for surveys of the Gulf of Maine. However, surveys have been regularly funded though the Scallop Research Set-Aside Program. The results of these surveys are presented in the Gulf of Maine appendix. The SSWG final report is available at this page: https://www.nefmc.org/committees/scallop-survey-working-group

- Continued survey data collection and analyses of scallop populations in the Gulf of Maine to support the future development of region specific reference points and growth estimates.
    - The Scallop RSA continues to support annual surveys in the Gulf of Maine. A working paper was prepared as part of this assessment to summarize the state of survey information.

Gonad-based estimates of SSB and reference points:

- Further work to develop gonad based estimates of SSB and reference points.(2020 management track assessment)
    - Gonad-based estimates were explored in this research track in the Mid-Atlantic SYM model. Ultimately, the working group did not recommend transitioning to this approach because the data was very noisy - would require more work to use to develop a Mid-Atlantic shell height to gonad weight relationship.

Scallop Forecasting Model (Scallop Area Management Simulator or SAMS), LPUE, and VMS data:

- Further refine and test methods for forecasting LPUE. (2018 Benchmark Assessment)
    - Limited progress, and further review of LPUE models is needed. Some development of spatial choice models for scallop fishermen, which have a role in LPUE forecasts. There have also been some changes to the LPUE model in annual management actions.
- Develop a spatially-explicit methodology for forecasting the abundance and distribution of sea scallops by incorporating spatial data from surveys, landings, and fleet effort (aka GEOSAMS). (2018 Benchmark Assessment)
    - Progress has been made. The working group received a presentation on the development of GEOSAMS in January. Funding for this work was halted due to a budget cut that supported a part-time statistical programmer.

- Revive and streamline previously-developed methods for interpreting VMS data. (2018 Benchmark Assessment)
    - Limited progress. Scallop Plan Development has developed a standard way to categorize fishing activity using speed filters. The simple speed cutoff can mask fishing and/or steaming. Using a depth cutoff would help, and the working group noted that there may be more sophisticated methods to explore, such as a Hidden Markov model.
- The SAMS model seems to be having some difficulty capturing some of the recent stock changes. Recommendation to look into new treatments for recruitment assumptions in the SAMS model. A revamp of the SAMS model to allow for more spatial estimation would be another fruitful area to explore. The SSC recommends a review of the SAMS model in the next management track assessment, and supports NEFSC's development of a geostatistical SAMS model for the 2024 research track assessment. (2020 SSC Report)
    - No management track or formal review of SAMS has occurred. The working group anticipates that TOR 6 will be partially fulfilled.
    - One of the performance problems in SAMS has been overestimation. In the CASA and SYM models presented by the working group, estimates of M have increased in this assessment. These changes can be carried through in the SAMS model.
    - Downward adjustments to recruitment assumptions in SAMS may also improve performance in aggregate for the model. The working group discussed the practical importance of tracking and reviewing the performance of forecasts for individual areas, since these are used in making fishery allocations.
- The SSC recommends that ongoing research on potential drivers of changes in sea scallop stock dynamics (e.g., changing ocean conditions, including ocean acidification and warming) be included in the upcoming review of the SAMS model and in the 2024 research track assessment for scallops. (2021 SSC Report)
    - New sources of information include the working paper on thermal conditions (TOR 1), and revised CASA models (TOR 4).
    - There is ongoing work at the NEFSC's Milford Lab on ocean acidification.
    - Future work is needed to incorporate this work into SAMS and reviewing the results of the forecasts.
- The SSC recommends continued evaluation of the performance of the projection model and the need for a more holistic evaluation of changes in stock dynamics, a synoptic evaluation of potential drivers (e.g., changing ocean conditions,

including ocean warming), and revision of model assumptions to account for these changes. (2022 SSC Report)

  - See previous responses.
  - Some annual ad-hoc adjustments to M were made for management purposes in estimation areas at the southern end of the range (Virginia Beach, Delmarva). A retrospective-type analysis on historical area-specific SAMS projections is recommended; a quantitative measure of past bias could be used as a tool to inform future projections. (2024 SSC Report)
  - Evaluation of rotational management has some. Summary of SAMS performance over time is shown in Figure 4.2 in TOR 6. Future work can focus on plots by sub-areas, possibly later this year.

<u>Environmental Drivers and Stock Dynamics:</u>

- Continued investigation of discard mortality, particularly during warm water periods, by incorporating environmental data. (2018 Benchmark Assessment)
  - Thermal mortality has not yet been adequately addressed - still assuming 20% discard mortality but probably too low in Mid-Atlantic because of water temperatures. Discard mortality remains an uncertainty, and further work is needed.
- The SSC recommends investigation of the environmental drivers affecting the productivity of this stock, such as those influencing recruitment and natural mortality. (2022 SSC Report)
- Research on the implications of expected future increases in bottom temperature on natural mortality and how these impacts on scallop survival will likely continue to extend northward. The SSC suggests that a thorough examination of the impacts of ecosystem and climate change on population dynamics is critical during the ongoing Sea Scallop Research Track Stock Assessment. (2023 SSC Report)
- The SSC supports continued monitoring of changes in the dynamics of this stock (i.e., recruitment, growth, and natural mortality) and research to understand the role of environmental drivers as this represents the most pressing concern facing the future of this fishery. (2024 SSC Report)

  - These three recommendations are partially addressed in TOR 1, however additional work is needed. The working group noted the importance of integrating changing environmental and oceanographic information into scallop science and management, and the need for cross-branch collaboration with the Ecosystem Dynamics and Assessment Branch

(EDAB). Additional collaborative and trans-disciplinary research programs are needed to advance this work.

Stock Assessment Model, Aging, Growth, Selectivity:

- Analyze past juvenile scallop mortality events and develop better methods to model time-varying mortality in the assessment models. (2018 Benchmark Assessment)
    - Continued work on this topic in this research track assessment (TOR 4). The CASA models include age/time varying estimates of natural mortality.
- Continue development of scallop ageing methods and examination of scallop growth processes including density dependent effects. (2018 Benchmark Assessment)
    - Peer reviewed work in this area: Kowaleski, KR & Roman, SA & Mann, R & Rudders, DB. (2024). Extreme population densities reduce reproductive effort of Atlantic sea scallops in high-density recruitment events. Marine Ecology Progress Series. 746. 10.3354/meps14688.
    - More work should be conducted to examine the estimation methods of growth transition matrices, enabling these matrices to better capture spatial and temporal changes in growth so that these changes can be incorporated into CASA models.
- Investigate methods to better estimating biomass and abundance variances from Hab-cam optical surveys including development of Bayesian geostatistical methods. (2018 Benchmark Assessment)
    - Peer-reviewed paper on this topic: Duskey, Elizabeth & Hart, Dvora & Chang, J.-H & Sullivan, P.J.. (2023). Partitioning spatial dynamics in abundance of marine fisheries stocks between fine- and broad-scale variation: A Bayesian approach. Fisheries Research. 267. 106816. 10.1016/j.fishres.2023.106816.
    - Progress has been made in this area, but it is not ready for implementation.
- Investigate and estimate current and historical unreported landings and effects of spatially heterogeneous fishing mortality on mortality estimates. (2018 Benchmark Assessment)
    - No recent progress on this topic. Unreported fishing mortality early in the assessment time series is suspected, though there are no obvious solutions to address the challenge. Peer-reviewed work from 2023: Hart, Dvora. (2003). Yield- and biomass-per-recruit analysis for rotational fisheries, with an application to the Atlantic sea scallop (Placopecten magellanicus). Fishery Bulletin. 101. 44-57.

- Hart DR (2001) Individual-based yield-per-recruit analysis, with an application to the Atlantic sea scallop, *Placopecten magellanicus*. Can J Fish Aquat Sci 58: 2351-2358
- Investigate and parameterize sub-lethal effects of disease, parasites, or discarding on mortality, growth, and landings. (2018 Benchmark Assessment)
  - Ongoing work at the Virginia Institute of Marine Science looking at shell disease, and nematodes. Multiple survey groups are tracking scallop condition.
  - Further exploration conducted by researchers at VIMS, see: Rudders et. al. (2019). An Investigation into the Scallop Parasite Outbreak on the Mid-Atlantic Shelf: Transmission Pathways, Spatio-Temporal Variation of Infection and Consequences to Marketability. RSA Award Number: NA16NMF4540043. VIMS Marine Resource Report No. 2019-02.
- Continue improvements of observer recordings for vessel fishing behavior including deck loading and shucking dynamics in responses to disease or poor scallop health. (2018 Benchmark Assessment)
  - Observer protocols were modified a few years ago to better track scallop health and meat condition. If grey meats or parasites are present, observers must resample meat weight at least twice per watch and weigh affected meats separately from clean meats
  - Observers not currently checking for shell blister disease.
- Continue investigating the extent of incidental fishing mortality, particularly on hard bottom habitats. (2018 Benchmark Assessment)
  - Not complete. Research in this area by the Virginia Institute of Marine Science and University of Delaware in 2017. This work did not focus on hard bottom habitats. See more at: Ferraro, Danielle & Trembanis, Arthur & Miller, Douglas & Rudders, David. (2017). Estimates of Sea Scallop (Placopecten magellanicus) Incidental Mortality from Photographic Multiple Before—After-Control—Impact Surveys. Journal of Shellfish Research. 36. 615-626. 10.2983/035.036.0310.
- Continued development of the SYM model that models selectivity dynamically as a function of full recruitment fishing mortality. (2020 Management Track)
  - Not complete, and work is still underway at the NEFSC.
- Transitioning the CASA model into the "next generation" of assessments would also be a natural progression; a state-space model with parameters that can be connected to environmental variables could result in more accurate estimates of biomass and fishing mortality rate and reference points with an improved

understanding of uncertainty and the relationship between scallops and the ecosystem. (2024 SSC Report)

- o This recommendation was made while the research track process was ongoing. The working group recommended updating the CASA models for this research track at the start of this process in 2023. This SSC recommendation is carried forward in future research recommendations below, and in the working group's response to the TOR 7 on quantitative assessment of ecosystem influences. The working group notes that CASA is attempting to capture process error in time-varying natural mortality.
- o State-space models are being used for this scallop species in other regions. DFO Canada is using a delayed difference state-space model to assess the Atlantic sea scallop resource.

3. *Quantitative Assessment of Ecosystem Influences*

Ecosystem influences considered in TOR 1 were not incorporated into the CASA models. In 2024, the SSC recommended the consideration of a state-space assessment model. The SSC stated that a state-space model with parameters that can connect environmental variables could result in more accurate estimates of biomass and fishing mortality rate and reference points with an improved understanding of uncertainty and the relationship between scallops and the ecosystem. This recommendation has been carried forward in the Working Group's recommendations for future research.

While it is recognized that environmental drivers can be incorporated directly into stock assessments, there are many other avenues for informing scientific and management decisions using environmental data. The NEFMC's Risk Policy identifies climate and ecosystem considerations (climate vulnerability and fish condition) as being important in the development of management advice. These factors are evaluated and translated into a level of risk aversion for use in management and catch advice.

The working group noted the importance of ongoing support from the NEFSC's Ecosystem Dynamics and Assessment Branch to integrate changing environmental and oceanographic information into scallop science and management.

4. *New Research Recommendations for Future Research, Data Collection, and Assessment Methodology*

The working group recommends the following new research recommendations grouped by category. The recommendations are listed below in proposed order of priority.

Aging

- Age scallops that are available from the 1990s to fill data gaps.

Projection Methods

- Review the performance of area-specific forecasts (e.g. a specific SAMS estimation areas) over time. Consider the status of an area (open and being fished vs. closed) in this review. Reference earlier work complete as part of the Council's Evaluation of Rotational Management.
- Develop spatial strata (SAMS Areas) in the Gulf of Maine that can be used to direct future surveys, and would be the basis for future forecasting of abundance and biomass.
- Continue the development of new forecasting models, including GeoSAMS.

Assessment Models

- Further investigate the impacts of environmental impacts on scallops, and consider ways to incorporate these directly into stock assessment models.
- Develop a state-space assessment model with parameters that can be connected to environmental variables.

Growth

- Length based CASA models are dependent upon growth estimates. Research that investigates estimates of growth, and an exhaustive review of prior estimates of growth should be conducted.

Natural Mortality

- Explore and refine predator-prey interaction models to improve natural mortality estimates. Investigate changes in M due to increases in sea star predation.

Stock Structure

- Examine current assumptions about scallop stock structure in the Gulf of Maine, Georges Bank, and the Mid-Atlantic.

Environmental Change

- Investigate the implications of warming sea surface temperatures and the reduced optimal thermal window on larval survival to refine models of density-independent mortality.

- Research to characterize the interplay of food availability, oxygen, and temperature on natural mortality.
- Investigate how ocean forecasting products perform for scallop usability in assessment and forecasting models.

Survey Data

- Continue to analyze the use of artificial intelligence (AI) algorithms in annotations from images taken using optical survey tools (HabCam and Drop Camera).
- Continue to survey and analyze scallops in the Gulf of Maine.